

GOURE, Niger

WMO: 610450

Lat: 13.983N

Lon: 10.300E

Elev: 460

StdP: 95.92

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.6	13.8	-8.2	2.0	25.9	-8.1	2.0	26.4	9.1	21.7	8.2	20.7	1.0	40	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.4	42.2	19.9	41.3	19.8	40.4	19.8	25.7	34.0	25.2	33.1	24.8	32.5	2.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.6	28.5	23.2	19.0	28.3	22.8	18.6	28.2	81.8	33.9	79.7	33.2	78.1	32.4	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.2	5.2	DB	10.0	43.9	1.0	0.6	9.3	44.4	8.7	44.7	8.1	45.1	7.4	45.5	
			WB	3.6	27.5	0.7	1.8	3.2	28.7	2.8	29.8	2.4	30.8	1.9	32.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	21.6	24.8	29.2	32.8	34.0	33.1	30.3	28.5	30.1	30.4	26.5	22.4
	DBStd	4.51	2.82	3.28	2.61	2.04	1.73	2.19	2.06	1.89	1.88	1.61	2.26	2.33
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	3	3	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6805	359	416	594	683	744	692	629	574	602	633	494	384
	CDD18.3	3767	103	182	336	433	486	442	371	316	352	374	244	127
	CDH23.3	51093	1073	2083	4588	6733	7813	6700	4898	3618	4447	5078	2826	1236
CDH26.7	30339	427	1065	2781	4534	5401	4378	2682	1679	2401	3008	1509	474	
Wind	WSAvg	1.8	2.1	1.9	1.8	1.7	2.0	2.6	2.4	1.6	1.1	1.3	1.6	1.8
Precipitation	PrecAvg	282	0	0	0	1	8	25	94	120	48	4	0	0
	PrecMax	497	0	0	3	26	35	102	235	291	156	40	0	0
	PrecMin	90	0	0	0	0	0	0	18	6	3	0	0	0
	PrecStd	95	0	0	0	5	9	23	47	59	39	9	0	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	38.4	40.8	43.0	43.4	42.2	39.6	36.8	39.4	39.9	37.7	35.2
		MCWB	15.8	17.4	17.4	19.7	20.2	21.4	22.0	24.6	20.4	19.6	18.3	16.3
	2%	DB	33.2	36.7	39.6	42.0	42.4	41.1	38.0	35.5	38.3	38.9	36.6	32.9
		MCWB	14.7	16.2	17.5	19.4	20.0	21.4	22.9	24.6	21.3	19.5	17.7	15.5
	5%	DB	31.3	34.8	38.4	41.1	41.6	40.1	36.5	34.3	36.9	38.0	35.3	31.2
		MCWB	14.0	15.6	17.0	19.0	19.9	21.8	23.3	24.5	22.4	19.5	16.9	14.7
	10%	DB	29.2	32.8	36.9	40.1	40.6	38.9	35.2	33.2	35.6	37.0	33.8	29.4
		MCWB	13.1	14.5	16.3	18.8	20.0	22.2	23.5	24.3	23.0	19.2	16.1	13.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.2	19.6	21.2	26.2	25.7	26.1	26.0	26.1	25.9	25.3	20.5	17.7
		MCDB	32.9	33.6	37.9	39.9	38.7	36.1	34.0	33.1	33.6	33.2	34.8	31.6
	2%	WB	15.4	17.1	19.5	23.4	24.2	24.9	25.1	25.5	25.4	24.2	18.7	16.4
		MCDB	31.8	34.7	35.7	37.0	36.3	35.1	32.9	32.5	32.9	32.4	34.8	31.0
	5%	WB	14.3	15.9	17.9	21.8	23.6	24.3	24.7	25.1	25.0	23.4	17.5	15.3
		MCDB	30.2	33.3	35.8	35.6	35.5	34.6	32.4	32.0	32.4	31.6	33.9	29.5
	10%	WB	13.2	14.9	16.9	20.4	22.9	23.8	24.2	24.7	24.6	22.5	16.5	14.2
		MCDB	28.5	32.2	35.5	35.7	34.8	34.0	31.9	31.3	31.7	30.9	32.7	28.5
Mean Daily Temperature Range	5% DB	MDBR	14.5	14.7	14.7	14.5	13.4	11.4	10.0	9.2	11.1	13.3	14.5	14.2
		MCDBR	16.9	16.6	16.4	15.3	14.6	12.6	11.3	10.5	12.8	14.3	15.6	15.9
		MCWBR	7.1	6.5	6.3	6.0	5.3	4.5	3.1	3.0	4.1	4.8	6.1	6.7
	5% WB	MCDBR	16.4	16.2	15.1	12.7	12.0	11.1	10.0	9.7	10.4	11.8	15.1	15.4
		MCWBR	7.3	6.8	6.6	6.4	4.6	3.9	3.3	3.3	3.7	4.5	6.5	7.1
Clear-Sky Solar Irradiance	taub		0.570	0.626	0.679	0.721	0.708	0.705	0.663	0.622	0.629	0.660	0.567	0.559
	taud		1.673	1.548	1.462	1.423	1.461	1.521	1.652	1.736	1.698	1.567	1.660	1.687
	Ebn at Noon		727	709	687	658	655	649	680	717	713	674	723	721
	Edh at Noon		240	282	314	326	309	287	253	235	245	273	240	231
All-Sky Solar Radiation	RadAvg		5.65	6.12	6.54	6.71	6.72	6.41	6.11	6.01	6.31	6.12	5.78	5.46
	RadStd		0.17	0.13	0.29	0.23	0.26	0.26	0.16	0.21	0.14	0.19	0.16	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page